

PHASE II : Forensic Acquisition Report

1. Target System Description

For this phase of the project, I created a local test environment using Kali Linux running on a virtual machine. Since I am working within a controlled environment, I simulated a forensic case by creating a suspicious ZIP file that represents unknown or potentially risky data found on a system.

The purpose of this setup was to create a realistic but safe scenario where I could practice forensic data acquisition techniques without affecting a real system. The simulated scenario involves a compressed file that was discovered with no clear origin, containing multiple files that needed to be examined for hidden or sensitive information.

2. Acquisition Plan

The main objective of this phase was to acquire and examine the contents of a suspicious ZIP file. Instead of performing full disk imaging or memory capture, I focused on file-level acquisition and analysis, which is still a valid forensic approach in situations where specific files are under investigation.

The tools selected for this phase include:

- zip / unzip (Kali Linux tools) – used to compress and extract files

- ls command – used to identify files, including hidden ones
- cat command – used to view file contents

These tools were chosen because they are simple, effective, and already built into Kali Linux, making them practical for quick forensic analysis.

3. Step-by-Step Acquisition Procedures

3.1 Creating the Evidence

To simulate a real-world case, I created a folder containing multiple files, including text files and a hidden file. This was done using the following commands:

```
mkdir evidence_case  
cd evidence_case  
echo "Nothing important here" > notes.txt  
echo "password=admin123" > creds.txt  
touch image1.jpg  
touch log_file.log  
echo "hidden data" > .secret.txt
```

This created a mix of visible and hidden files, which is common in forensic investigations.

3.2 Creating the Suspicious ZIP File

The folder was then compressed into a ZIP file to simulate a suspicious archive:

```
cd ..  
zip -r suspicious.zip evidence_case
```

This file represents the main piece of evidence being investigated.

3.3 Listing ZIP Contents

Before extracting the file, I examined its contents using:

```
unzip -l suspicious.zip
```

This step allows investigators to see what is inside the archive without modifying it.

3.4 Extracting the ZIP File

The ZIP file was extracted using:

```
unzip suspicious.zip
```

This made all files available for further analysis.

3.5 Identifying Hidden Files

To check for hidden data, I used:

```
ls -la evidence_case
```

This revealed the presence of a hidden file (.secret.txt), which would not be visible with a normal directory listing.

3.6 Examining File Contents

Finally, I reviewed the contents of one of the files:

```
cat evidence_case/creds.txt
```

This revealed stored credentials, which could represent sensitive information in a real investigation.

4. Challenges and Mitigation

During this phase, there were no major technical issues, but one challenge was ensuring that the scenario remained realistic while still being simple to execute. This was addressed by creating a mix of visible and hidden files to simulate real-world forensic conditions.

Another consideration was maintaining proper structure and documentation so that each step could be clearly explained and repeated if needed.

5. Chain of Custody Documentation

- Evidence ID: ZIP-001
- Date/Time of Acquisition: (Insert your actual time)
- Description: Suspicious ZIP file containing multiple files
- Integrity Verification: Not applicable in this simulation, but in a real case, hashing (MD5/SHA256) would be used
- Storage Location: Local Kali Linux virtual machine

6. Summary and Reflection

This phase helped me understand how forensic data acquisition can be applied even at a file level. By creating and analyzing a suspicious ZIP file, I was able to simulate a real investigation scenario and practice identifying hidden files and sensitive data.

The process was straightforward, but it highlighted the importance of careful documentation and attention to detail. This experience prepares me for Phase 3, where I will perform deeper analysis on the collected data and look for patterns or indicators of suspicious activity.